

MATH 3191 Written Homework 2
____/20 points

Name: _____

Due: February 2nd

1. (5 points) Write the solution set to the following system of equations in vector (parametric) form

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 6 & 10 \\ 2 & 7 & -12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 12 \\ 13 \end{bmatrix}$$

2. (15 total points) Answer the following questions about the matrix associated with a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$

$$T\left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 6 \\ 2 \end{bmatrix}, T\left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}, T\left(\begin{bmatrix} 0 \\ 2 \\ 5 \end{bmatrix}\right) = \begin{bmatrix} -3 \\ 7 \\ 3 \end{bmatrix}$$

- (a) (10 points) Compute a matrix $A \in \mathbb{R}^{3 \times 3}$ such that $T(\mathbf{x}) = A\mathbf{x}$

- (b) (5 points) Is T an invertible linear transformation? Why or why not?

Note: You might want to wait until after lecture slides 11 before trying this!